

Introduction & Objective:

This project aims to design and implement a functional AM receiver capable of tuning into Bangladesh Betar stations. The receiver combines RF signal detection, amplification and audio output using analog components. The objectives include:

- Applying AM communication theory in hardware design.
- Building and testing a working receiver circuit.
- Demonstrating noise mitigation techniques.
- Promoting teamwork and hands-on learning.

Components Used:

- Antenna (simple wire antenna)
- Variable capacitor (400pf)
- Inductor (for LC tuning circuit)
- Capacitors: 0.01 μ F, 0.1 μ F, 10 μ F, 100 μ F, 220 μ F
- Resistors (10 k Ω ,180 k Ω ,10 Ω)
- Diode
- Transistor (C828)HA MA ReceLg
- LM386 amplifier IC
- Speaker (1W)
- 9V battery

The circuit diagram shows a 100 kHz sine wave generator. It features an LM386 op-amp configured as a voltage follower. The non-inverting input (+) is connected to the output of a 2N3904 transistor. The transistor's base is biased by a voltage divider consisting of R1 (10k) and R2 (100k) connected to a +9V supply. The emitter of the transistor is connected to ground through R3 (10Ω). The collector is connected to a load ZL through a 10kΩ resistor. The op-amp's output is connected to the collector of the transistor. The op-amp is powered by a +9V supply and has a bypass capacitor C5 (100μF) connected to its ground pin. The input of the op-amp is connected to the output of the transistor through a 10kΩ resistor. The output of the op-amp is connected to the collector of the transistor through a 10kΩ resistor. The circuit also includes several other components: C1 (0.1μF) and C2 (100μF) are connected to the +9V supply; C3 (100k) is connected to the base of the transistor; C4 (200μF) is connected to the output of the op-amp; C6 (22pF) and C7 (400pF) are connected to the input of the op-amp; C8 (0.1μF) is connected to the emitter of the transistor; C9 (100μF) and C10 (0.04μF) are connected to the output of the op-amp. A transformer is connected to the input of the op-amp, with its primary winding connected to the +9V supply and its secondary winding connected to the input of the op-amp.

The AM receiver circuit is designed to operate within the standard amplitude modulation (AM) broadcast band, typically ranging from 530 kHz to 1600 kHz. The circuit employs a tuned LC resonant network, consisting of an inductor (L1) and capacitors (C4, C5, C6). This network acts as a bandpass filter, allowing only the desired frequency to pass while attenuating others. This analog tuning method is simple yet effective, enabling selective reception without digital control.

Circuit working procedure:

Antenna(A1): The antenna captures the AM radio signal from the air. Here we want catch 1.2 MHz frequency for radio station Bangladesh Betar. We used the LC circuit that selects the desired station by tuning it to the frequency of the station.

$$F = \frac{1}{2\pi\sqrt{L(C7)}}$$

Here, $F = 1.2\text{MHz}$, $C7 = 400\text{pt}$.

SO, $L = 44\text{UH}$

L is a coil wound on a ferrite rod with 60 turns we used. cu wire diameter is 0.8mm.

D1(1N4148): The diode removes the carrier and leaves the audio part only.

Transistor(C828): The C828 amplifies this weak signal to a level suitable for further amplification by the LM386.

Variable Resistor(10K): Adjust the gain or volume of the LM386 amplifier.

LM386: This will allow the radio for good earth connection and LM386 provides enough amplification power to drive a small speaker.

Speaker(8KW): The speaker converts the electrical audio signal into sound wave.

Capacitors: The capacitors make sure only the AC signal goes through and not DC.

Resistor: The resistors around the transistors used for the biasing of the transistors.

Battery(9V):power up the circuit.

Procedure:

We started by setting up the LC circuit using the inductor and the variable capacitor, and connected the antenna to it so that it could pick up the AM signals. Then we placed the diode as the detector to separate the audio from the carrier. Once this part was working, we added the transistor to make the signal stronger since the detected audio was very weak. The LM386 amplifier was connected with the necessary capacitors and resistors, and we linked its output to a small speaker. Finally we connected a 9V battery to power the circuit and tried tuning the variable capacitor around the medium-wave AM band, close to 1.2MHz.

Noise Cancellation:

- Capacitors C5 and C9 set the overall gain of the LM386.
- Resistor R3 and capacitor C10 prevent unwanted instability of the LM386's output by providing a known load at frequencies above audio. Its a RC circuit.
- Use Bypass capacitor.

Applications:

It's beneficial for the beginner level radio project. Doing this project we understood the basics of LC circuit and how it can help us tune to our desired radio frequency range, because of their specific resonance at the specific combination. We also got the idea of the noise cancellation and how we can reduce the noise by using filter by removing mostly the very high frequency. Also understood how the capacitors can remove the unwanted DC signal and keep the AC only. It demonstrates the basics of wireless communication and can be the beginning for the more advanced receiver designs.

Conclusion:

We successfully built an AM radio receiver using an LM386 amplifier. The project gave us experience for radio circuits and showed how signals are selected, detected, and amplified. Even though we could not perfectly lock at 1MHz, the experiment was useful for understanding the practical difficulties of tuning and signal reception.

Discussion:

When we tested the receiver, we could pick up signals, but tuning to exactly 1.2 MHz was very difficult. Even a small change in capacitor or coil value shifted the tuning frequency. Because we used normal components, the accuracy was not great. Another problem was noise and interference from the surroundings, which made the sound not very clear at times. Still we got the speaker to make some sound from the radio frequency.

